CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

8.4 The Diffraction Grating

Syllabus Section: 8. Superposition **Paper Type:** Multiple Choice (Paper 1) **Total:** 16 Questions (16 Marks)

Question 1 Easy [1 mark]

A diffraction pattern is observed when monochromatic light is incident on a diffraction grating.

Which row describes the effect of replacing the grating with one that has more lines per metre?

	Number of orders of diffraction visible	Angle between first and second orders of diffraction
A	increases	increases
B	increases	decreases
C	decreases	increases
D	decreases	decreases

A

B

C

D

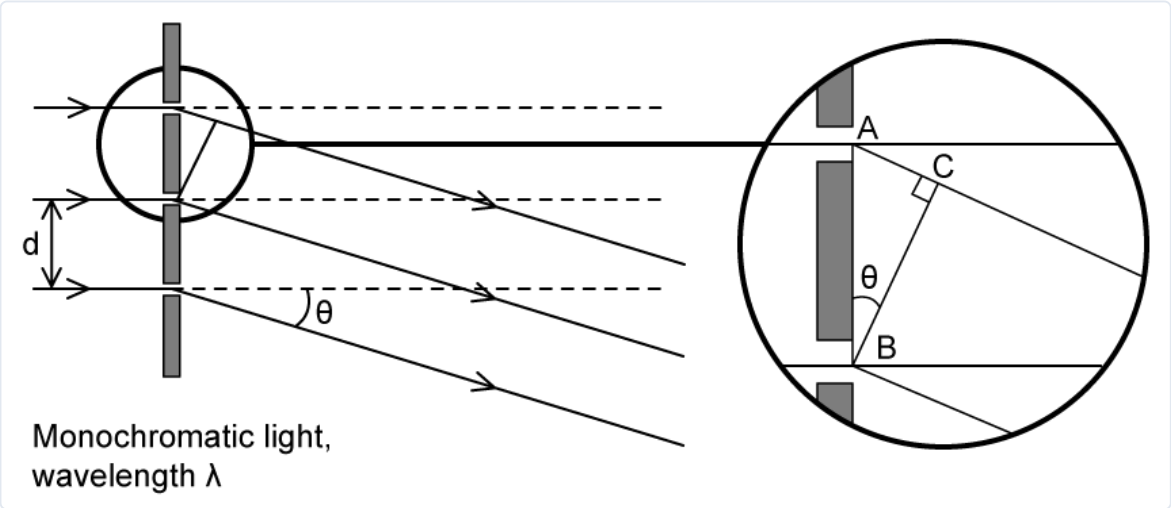
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The diffraction grating equation is given by
  - $d \ \ sin \ \ \theta = n\lambda$
- Changing the grating to one with more lines per metre causes the slit separation *d*, to decrease
- As the number of order, so diffraction visible decreases the angle between the first order and the second orders of diffraction increases
  - $n = \frac{d \ \ sin \ \ \theta}{\lambda}$ as *d* is decreasing
  - $\theta = sin^{-1} \frac{n\lambda}{d}$, this can only be between 0 and 90˚, and the angle increases

Question 2 Easy [1 mark]

The diagram shows a diffraction grating and a monochromatic source of light with wavelength (λ).



What happens to the angle of diffraction (θ) when the wavelength of light is increased, but all the other equipment remains the same?

A The angle of diffraction remains the same

B The angle of diffraction is decreased

C The angle of diffraction is increased

D There is not enough information to know what will happen

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- When the wavelength of the light is increased, it must diffract through a gap that is much smaller in relation to its length
- This means it "bends" or diffracts more, creating a larger angle of diffraction (θ)
- This can also be seen from the diffraction grating equation
  - $d \sin(\theta) = n\lambda$
- $\sin(\theta)$ is directly proportional to λ
  - This means that if λ increases then so does $\sin(\theta)$
  - If $\sin(\theta)$ is increasing then so is θ

A is incorrect as the angle of diffraction increases, so does not remain the same.

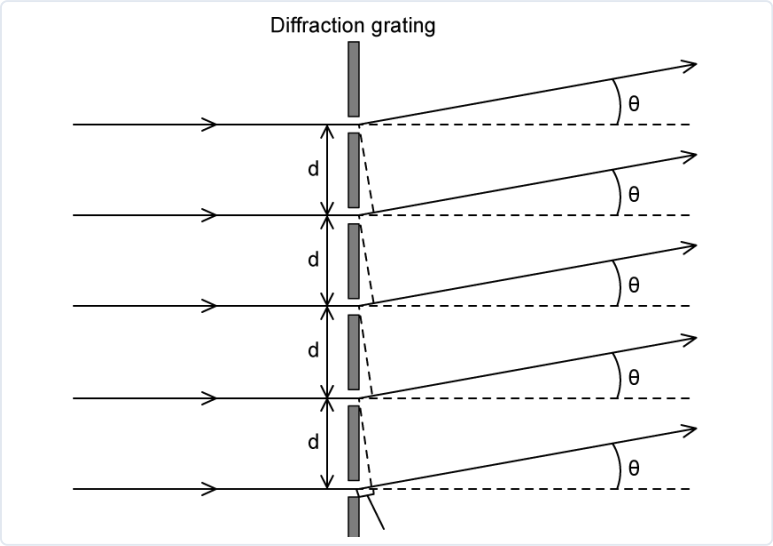
B is incorrect as the angle of diffraction increases, so does not decrease.

D is incorrect as there is enough information to know that the angle of diffraction will increase because of an increase in wavelength (λ).

It is important to be confident in how the terms of the diffraction equation ($d \sin(\theta) = n\lambda$) relate to each other. This is the easiest method to answer questions where you are asked how one variable varies with another – always look for the equations linking them both!

Question 3 Easy [1 mark]

The diagram shows a diffraction grating with a slit distance (d) and a monochromatic source of light incident on the grating.



What happens to the angle of diffraction when the distance between the slits on the grating is reduced?

A The angle of diffraction remains the same

B The angle of diffraction is decreased

C The angle of diffraction is increased

D There is not enough information to know what will happen

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- We can use the diffraction grating equation to help understand this
  - $\frac{n\lambda}{d} = \sin(\theta)$
- This means that $\sin(\theta)$ is inversely proportional to d
  - $\sin(\theta) \propto \frac{1}{d}$
- When the slit separation (d) is decreased then $\sin(\theta)$ will increase
- As the angle of diffraction increases, $\sin(\theta)$ also increases
  - This can be seen from the graph of $\sin(\theta)$ between $0 < \theta < 90°\sin(\theta)$

- Until 90º, as θ increases so does sin(θ)

A is incorrect as the angle of diffraction increases, so does not remain the same.

B is incorrect as the angle of diffraction increases, so does not decrease.

D is incorrect as there is enough information to know that the angle of diffraction will increase because of a decrease in slit separation (d).

It is helpful to be able to sketch graphs of sin(θ) and know how it's size changes with different values of θ, this is not always the case for the other trigonometric functions! As θ increases, so does sin(θ) only for **certain values of θ.**

Question 4 Easy [1 mark]

Monochromatic light incident on a diffraction grating creates both primary and secondary maxima in the diffraction pattern.

What is the relationship between the number of slits in the diffraction grating, when the number of slits is less than 20, and the number of secondary maxima that can be observed on the screen?

A The number of primary and secondary maxima in each diffraction pattern is the same

B The number of secondary maxima in the diffraction pattern is always one less than the number of primary maxima.

C The number of secondary maxima in the diffraction pattern is always the same as the number of slits in the diffraction grating

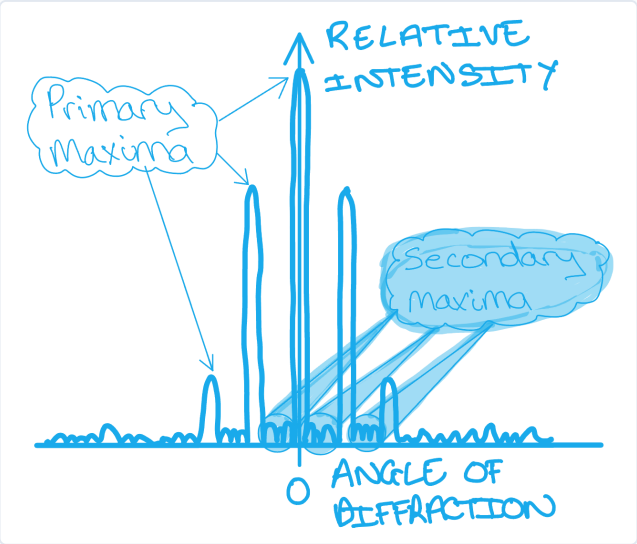
D The number of secondary maxima between the primary maxima is always two less than the number of slits in the diffraction grating

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- It can be observed by investigating light incident on a diffraction grating with different numbers of slits that:
  - When the number of slits is **greater** than two ($N > 2$) then **secondary** maxima appear
- E.g. when there are 3 slits in the diffraction grating then there is 3–2 = 1 secondary maxima between each primary maxima
- When there are 4 slits in the diffraction grating then there are 4 –2 = 2 secondary maxima between each primary maxima
  - Hence, the relationship between the number of secondary maxima and the number of slits in the diffraction grating is the **number of slits minus two**
- So, option **D** is correct

The diagram illustrates the difference between primary and secondary maxima. It shows the diffraction pattern for a diffraction grating with **five** slits. The number of secondary maxima visible is N − 2 = 5 − 2 = 3.


RELATIVE INTENSITY
Primary Maxima
Secondary Maxima
0
ANGLE OF DIFFRACTION

Question 5 Easy [1 mark]

What is a diffraction grating?

A Two slits created as two gaps within a material through which waves can be diffracted

B A plate consisting of a very large number of parallel, identical, closely-spaced slits

C A circular aperture through which light can be diffracted and then images resolved

D A photographic filter that allows light in only one plane to pass through

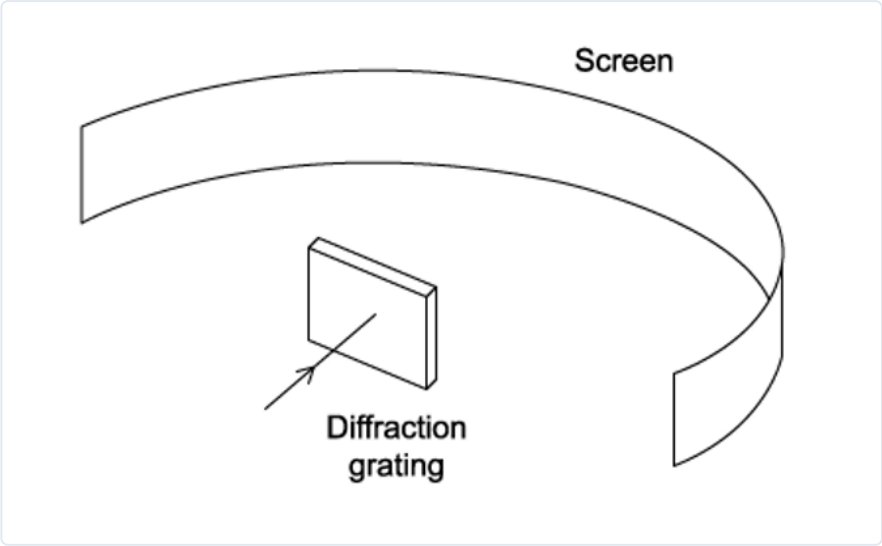
Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Diffraction gratings are made up of hundreds of slits per mm of their width
- The slits must all be:
  - In the same direction (therefore parallel to each other)
  - Have the same width and height (identical to each other)
  - Very close together
- This arrangement makes it possible to observe:
  - The diffraction of monochromatic visible light
  - The diffraction of visible white light into its colours
  - Phenomenons such as spectroscopy

Question 6 Medium [1 mark]

A beam of monochromatic light with a wavelength of 690 nm is directed at a diffraction grating with 300 lines per mm as shown in the diagram



This set up produces a series of maxima on the screen, what is the greatest number of maxima that can be observed?

A 9

B 8

C 5

D 4

Mark Scheme & Explanation:

*The correct answer is **A** because:*

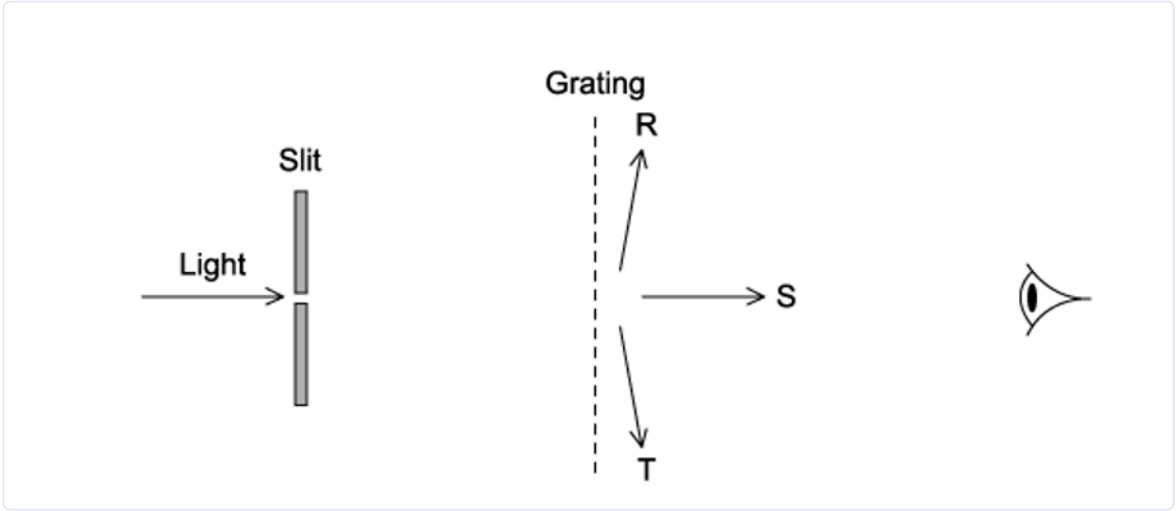
- The diffraction grating equation is given by
  - $d \quad sin \quad \theta = n\lambda$
- Where:
  - Slit separation $d = \frac{1}{300} = 3.33 \times 10^{-3} \quad \text{mm} = 3.33 \times 10^{-6} \quad \text{m}$
  - Wavelength $\lambda = 690 \times 10^{-9} \quad \text{m}$
  - Number of bright fringes on one side of the central maximum = n

- The maximum angle θ from the central fringe is 90° on each side
- Rearrange the diffraction equation to make n the subject:
  - $d \sin \theta = n\lambda$
  - $n = \frac{d \sin \theta}{\lambda} = \frac{3.33 \times 10^{-6} \sin 90°}{690 \times 10^{-9}} = 4.8$
- This means, the 4th bright fringe can be observed but not the 5th bright fringe
  - Since 4.8 < 5
- We will also have 4 bright fringes on the other side and one central fringe
  - Therefore, the total number of fringes = 4 + 1 + 4 = 9

Question 7 Medium [1 mark]

Monochromatic light with a wavelength of 600 nm is observed through a diffraction grating with 500 lines per mm.

The light is first passed through a narrow slit where the lines from the slit are parallel to the lines in the grating.



The observer changes position to; R – parallel with the grating, S – opposite the slit, T – parallel to the grating on the opposite side.

What is the total number of images they would see from R, S and T combined?

A 7

B 6

C 4

D 3

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The diffraction grating equation is given by
  - $d \ \ sin \ \ \theta = n\lambda$
- Where:
  - Slit separation $d = \frac{1}{500} = 2.0 \times 10^{-3} \ \ \text{mm} = 2.0 \times 10^{-6} \ \ \text{m}$
  - Wavelength $\lambda = 600 \times 10^{-9} \ \ \text{m}$
  - Number of images on one side = *n*

- The maximum angle θ from the central fringe is 90° on each side
- Rearranging for n :
  - $d \quad sin \quad \theta = n\lambda$
  - $n = \frac{d \quad sin \quad \theta}{\lambda} = \frac{2.0 \times 10^{-6} sin \quad 90°}{600 \times 10^{-9}} = 3.3$
- This means, the 3rd image can be seen but not the 4th image since we cannot have 3.3 images, so, we can have up to 3 images on one side
- We will also have 3 bright fringes on the other side and one central fringe
  - Therefore, the total number of fringes = 3 + 1 + 3 = 7

Question 8 Medium [1 mark]

A student set up an experiment with a diffraction grating. They put a narrow beam of monochromatic light incident along the normal. They found that the third-order diffracted beams are formed at angles of 45° to the original direction.

Which of the following gives the highest order of diffracted beam produced by this grating ?

A 6th

B 5th

C 4th

D 3rd

Mark Scheme & Explanation:

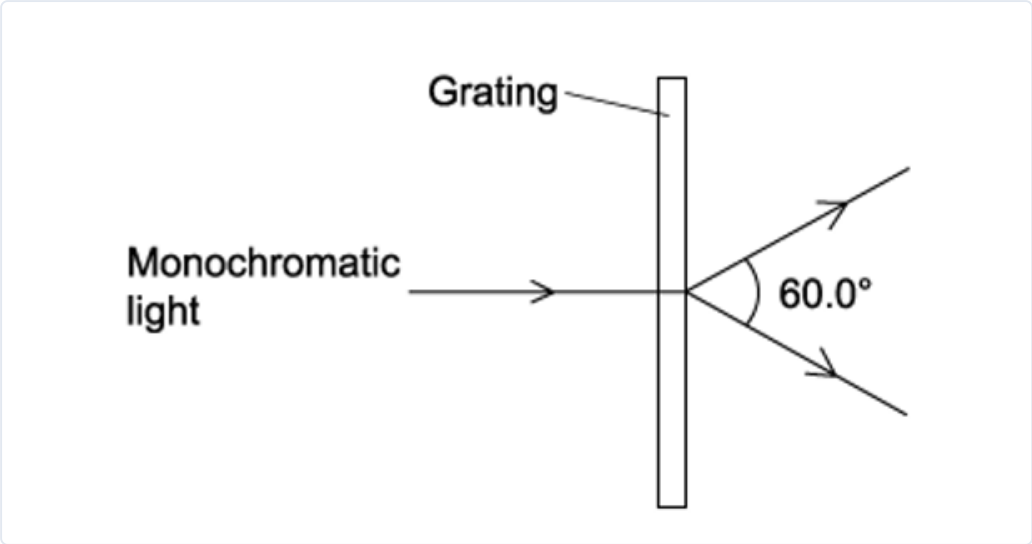
*The correct answer is **C** because:*

- The diffraction grating equation is given by
  - $d \ \ sin \ \ \theta = n\lambda$
- Rearrange the equation to obtain the ratio for n and $\sin\theta$
  - $\frac{\lambda}{d} = \frac{sin\theta}{n}$
- When n = 3 (third order), θ = 45°
  - $\frac{\lambda}{d} = \frac{sin45}{3}$
  - $\frac{\lambda}{d} = \frac{\sqrt{2}}{6}$
- For the highest order of diffracted beam, the angle θ should be equal to 90˚
- Consider the angle of 90°:
  - $\frac{\lambda}{d} = \frac{sin90}{n} = \frac{\sqrt{2}}{6}$
- Rearrange to obtain the highest order n
  - $\frac{\sqrt{2}}{6} = \frac{1}{n}$
  - $n = \frac{1}{\frac{\sqrt{2}}{6}} = \frac{6}{\sqrt{2}}$
  - $n = 4.2$

- The order should be an integer, so the highest order is $n = 4$

Question 9 Hard [1 mark]

The wavelength of monochromatic light can be determined using a diffraction grating. This is shown in the diagram.



The angle between the first-order diffraction maxima is 60°, and the spacing of the slits is 1.15×10^{-6} m.

What is the wavelength of the light?

A 575 nm

B 498 nm

C 996 nm

D 287 nm

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The diffraction grating equation is given by
  - $d \sin \theta = n\lambda$
- Where:
  - n = 1, since these are first-order lines
  - d is the split spacing = 1.15×10^{-6} m
  - θ is the angle between the light and the horizontal = $\frac{60^\circ}{2} = 30°$

- Rearranging for wavelength λ
  - $\lambda = \frac{d \ sin \ \theta}{n} = 1.15 \times 10^{-6} sin \ \ 30 = 5.75 \times 10^{-7} \ \ \text{m}$
- Therefore, the wavelength of the light is 575 nm

Question 10 Hard [1 mark]

When a parallel beam of white light is incident to normally on a diffraction grating the second-order and third-order spectra partially overlap.

When the second order wavelength is 600 nm, which of the following third-order wavelengths would appear at the same angle?

A 900 nm

B 600 nm

C 400 nm

D 300 nm

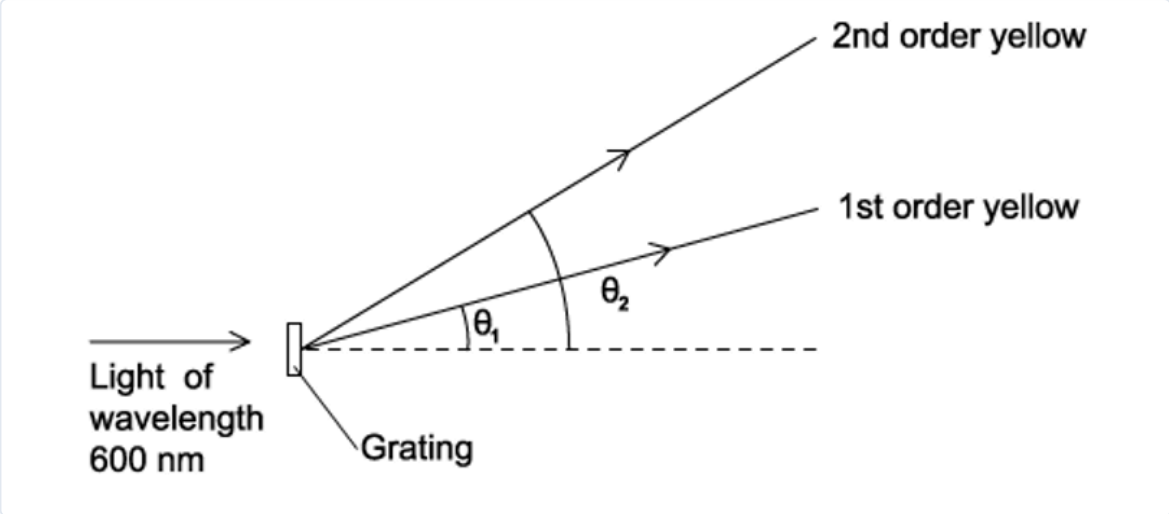
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The diffraction grating equation is given by
  - $d \ \ sin \ \ \theta_n = n\lambda_n$
- When the second-order and third-order spectra overlap:
  - $\theta_2 = \theta_3$
  - $d \ \ sin \ \ \theta_2 = 2\lambda_2 = d \ \ sin \ \ \theta_3 = 3\lambda_3$
  - Therefore, $2\lambda_2 = 3\lambda_3$
- Rearranging for λ_3:
  - $\lambda_3 = \frac{2}{3}\lambda_2 = \frac{2}{3} \times 600 = 400 \ \ \text{nm}$
- So, a wavelength of 400 nm would appear at the same angle as the 600 nm wavelength

Question 11 Hard [1 mark]

Yellow light with a wavelength of 600 nm is used in a diffraction grating experiment. The grating has a slit separation of 2.00 μm.



What would be the angular separation between the first order and second order maxima?

A 54.3°

B 36.9°

C 19.4°

D 17.5°

Mark Scheme & Explanation:

*The correct answer is **C** because:*

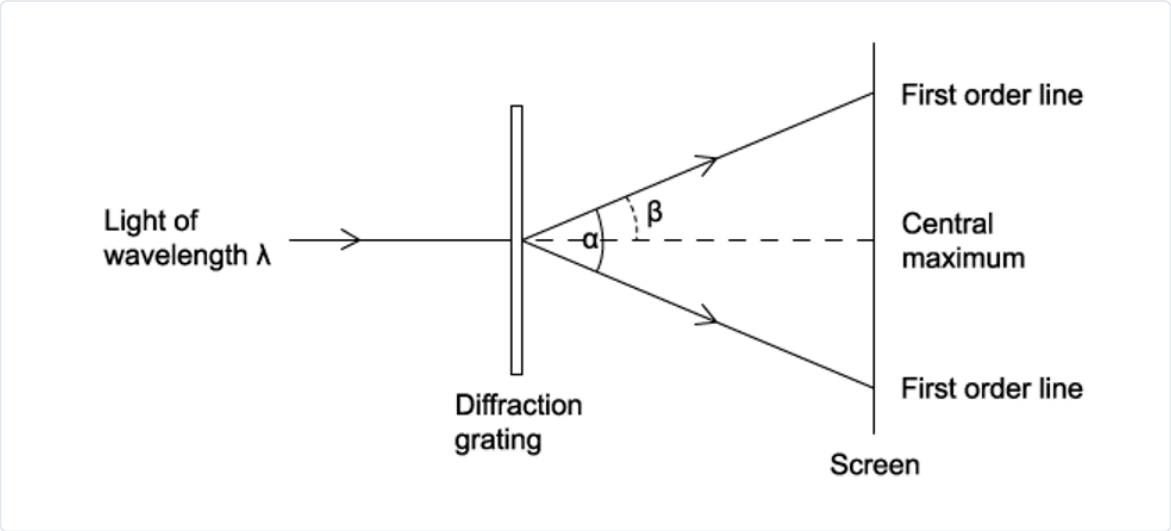
- The diffraction grating equation is given by
  - $d \ \ sin \ \ \theta = n\lambda$
- For n = 1
  - $d \ \ sin \ \ \theta_1 = n\lambda$
  - $sin \ \ \theta_1 = \frac{\lambda}{d} = \frac{600 \times 10^{-9}}{2.00 \times 10^{-6}} \ \ = \ \ 0.30$
  - So, $\theta_1 = sin^{-1} 0.30 = 17.45°$
- For n = 2

- $d \sin \theta_2 = 2\lambda$
- $\sin \theta_2 = \frac{2\lambda}{d} = 2 \times 0.30 = 0.60$
- So, $\theta_2 = \sin^{-1} 0.60 = 36.87°$

- Angular separation is equal to ($\theta_2 - \theta_1$), so:
  - $\theta_2 - \theta_1 = 36.87° - 17.45° = 19.42°$
- Therefore, the angular separation is equal to 19.4°

Question 12 Hard [1 mark]

An experiment was set up to investigate light passing through a diffraction grating with slit spacing *d*. The fringe pattern was observed on a screen.



Which of the following gives the angle between the two first-order lines?

A $2 \quad sin^{-1} \frac{\lambda}{2d}$

B $2 \quad \sin^{-1} \frac{\lambda}{d}$

C $\sin^{-1} \frac{\lambda}{2d}$

D $\sin^{-1} \frac{\lambda}{d}$

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The diffraction grating equation is given by
  - $d \quad sin \quad \theta = n\lambda$
- For $n = 1$
  - $sin \quad \beta = \frac{\lambda}{d}$
  - So, $\beta = sin^{-1} \frac{\lambda}{d}$
- From the diagram, $2\beta = \alpha$

- Therefore, $2\beta = \alpha = 2 \quad sin^{-1}\frac{\lambda}{d}$

Question 13 Hard [1 mark]

A student carried out an experiment on monochromatic light with a wavelength of 5.30×10^{-7} m. They set up the source so that it was incident normally to the diffraction grating.

The student observed that the first-order maximum was 15.4° to the direction of the incident light.

What would be the angle between the first order and second order maxima?

A 7.6°

B 15.4°

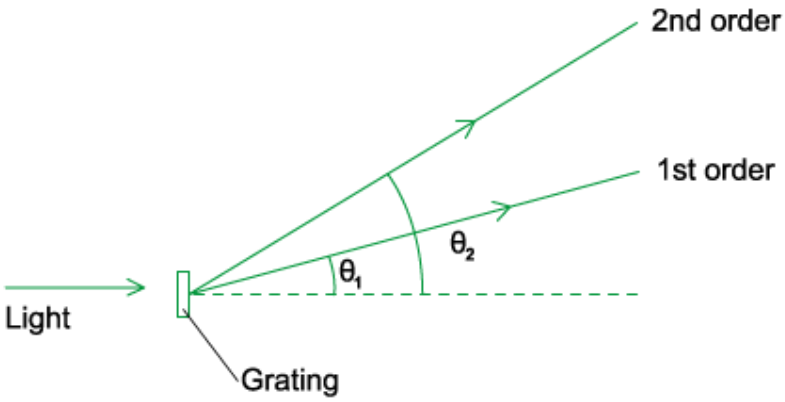
C 16.7°

D 32.1°

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The diffraction grating equation is given by
  - $d \ \ sin \ \ \theta = n\lambda$
- When n = 1
  - $sin \ \ \theta_1 = \frac{\lambda}{d}$
  - The angle θ_1 = 15.4°, so, $sin \ \ 15.4 = 0.2656$
- When n = 2
  - The angle is θ_2
  - $sin \ \ \theta_2 = \frac{2\lambda}{d}$
- $\frac{\lambda}{d}$ is the same at both the first-order and second-order maxima, so:
  - $\frac{\lambda}{d} = sin \ \ \theta_1 = \frac{1}{2} sin \ \ \theta_2 = 0.2656$
  - $sin \ \ \theta_2 = 2 \times 0.2656 = 0.5311$
  - $\theta_2 = \sin^{-1} 0.5311 = 32.08^{\circ}$
- Therefore, the angle between the first order and second order maxima is:
  - $\theta_2 - \theta_1 = 32.1° - 15.4° = 16.7°$


2nd order
1st order
θ2
θ1
Light
Grating

Question 14 Hard [1 mark]

A monochromatic light beam is incident normally to a diffraction grating and nine maximas are observed on the screen.

What is the range for the angle θ between the central and first-order maxima?

A $5.74° \leq \theta \leq 6.38°$

B $9.59° < \theta \leq 11.5°$

C $11.5° < \theta \leq 14.5°$

D $11.5° < \theta < 53.1°$

Mark Scheme & Explanation:

The correct answer is ***C*** *because*

- The diffraction grating equation is given by
  - $d \sin \theta = n\lambda$
- The maximum angle θ a fringe can be from the central maxima is 90°
- For the maximum value of n
  - $d \sin (90) = n\lambda$
  - $d = n\lambda$
  - $n = \frac{d}{\lambda}$
- There are nine maximas on the screen which means there are four maximas either side of the central maxima
- If the value of $\frac{d}{\lambda}$ is not a whole number, it must be rounded down as there cannot be a fraction of a fringe
- Therefore, $\frac{d}{\lambda}$is greater or equal to 4 and less than 5
  - $4 \leq n \leq 5$
  - $4 \leq \frac{d}{\lambda} < 5$
- When $n = 1$ for the first-order maxima
  - $d \sin \theta = \lambda$
  - $\sin \theta = \frac{\lambda}{d}$
  - $\theta = \sin^{-1}\frac{\lambda}{d}$
- If $\frac{d}{\lambda} = 4$

- $\frac{\lambda}{d} = \frac{1}{4}$
- $\theta = \sin^{-1}\frac{1}{4}$
- $\theta = 14.5°$ (3 s.f)

- If $\frac{d}{\lambda} = 5$
  - $\frac{\lambda}{d} = \frac{1}{5}$
  - $\theta = \sin^{-1}\frac{1}{5}$
  - $\theta = 11.5°$ (3 s.f)
- Therefore the range for the angle between the central and first-order maxima is
  - $11.5° < \theta \leq 14.5°$
- This is option **C**

Question 15 Hard [1 mark]

Fig. 1.1 shows a monochromatic light beam which is incident normally to a diffraction grating. The angle between the central and second-order maxima is 18°.

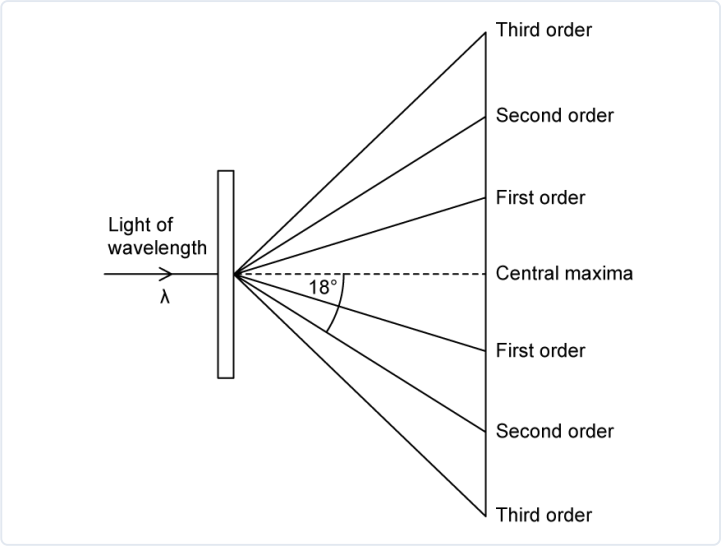


Fig. 1.1

What is the angle between the two third-order maxima lines?

A 27.6°

B 36.0°

C 54.0°

D 55.2°

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The known quantity is
  - The angle between the central and second-order maxima, θ = 18°
- The diffraction grating equation is given by
  - $d \ sin \ \theta = n\lambda$
- When n = 2 for the second-order maxima
  - $d \ sin 18 = 2\lambda$

- $\frac{\lambda}{d} = \frac{sin18}{2} = 0.1545$

- $\frac{\lambda}{d}$ remains constant for all of the maximas
- When n = 3 for the third-order maxima
  - $d \ sin \ \theta = 3\lambda$
  - $sin \ \theta = 3\frac{\lambda}{d} = 3 \times 0.1545$
  - $\theta = sin^{-1} \ 3 \times 0.1545 = 27.6°$(3 s.f)
- This is the angle between the central and third-order maxima
- The angle between the two third-order maximas is
  - $2 \times 27.6 = 55.2°$
- This is option **D**

Question 16 Hard [1 mark]

A physics student is investigating the properties of a diffraction grating using laser light with a wavelength of 6.20×10^{-7} m. The laser beam is directed perpendicularly onto the grating surface.

During the experiment, the student measures the second-order maximum at an angle of $32°$ from the normal to the grating.

What is the angle between the second-order and third-order maxima?

A 12°

B 21°

C 32°

D 53°

Mark Scheme & Explanation:

The correct answer is B

- The diffraction grating equation is given by

$$d \sin \theta = n\lambda$$

- When $n = 2$

$$\sin \theta_2 = \frac{2\lambda}{d}$$

- The angle between n = 2 and the normal is

$$\theta_2 = 32°$$

$$\sin 32° = 0.530$$

- When $n = 3$

$$\sin \theta_3 = \frac{3\lambda}{d}$$

- $\frac{\lambda}{d}$ is the same at both the second-order and third-order maxima, so:

$$\frac{\lambda}{d} = \frac{\sin \theta_2}{2} = \frac{\sin \theta_3}{3} = \frac{0.530}{2} = 0.265$$

$$\sin \theta_3 = 3 \times 0.265 = 0.795$$

$$\theta_3 = \sin^{-1}(0.795) = 52.7 \approx 53°$$

- Therefore, the angle between the second order and third order maxima is:

$$\theta_3 - \theta_2 = 53° - 32° = 21°$$